SECP3744 - UTHM CASE STUDY

UTHM INVENTORY SYSTEM PROJECT REPORT

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UTHM CASE STUDY
UTHM INVENTORY SYSTEM

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DECLARATION

I declare that this Choose an item. entitled *"title of the thesis"* is the result of my own research except as cited in the references. The Choose an item. has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

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ABSTRACT

Enterprise Architecture (EA) plays a crucial role in aligning information systems with organisational strategies, particularly within higher education institutions that rely heavily on integrated digital platforms. Universiti Tun Hussein Onn Malaysia (UTHM) has adopted EA principles to support its academic, administrative, and research operations through various subsystems. However, challenges related to system integration, interoperability, and architectural consistency continue to affect operational efficiency and user experience. This study aims to analyse and enhance the existing enterprise architecture framework at UTHM to better support institutional objectives.

The System Development Life Cycle (SDLC) methodology was employed to guide the planning, analysis, design, and implementation phases of the proposed architectural improvements. Data were collected through document analysis, system observation, and stakeholder input to identify current limitations and requirements. Based on the findings, an improved enterprise architecture model was designed to address identified gaps, focusing on better system integration, scalability, and governance.

The outcome of this study demonstrates that the application of a structured SDLC approach can effectively support the enhancement of enterprise architecture in a higher education environment. The proposed improvements contribute to improved system alignment, reduced redundancy, and more efficient service delivery for students and staff. This study provides valuable insights for higher education institutions seeking to strengthen their digital transformation initiatives through systematic enterprise architecture development.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

UTHM	-	Universiti Tun Hussein Onn Malaysia
EA	-	Enterprise Architecture
HEI	-	Higher Education Institutions
LMS	-	Learning Management System
SA	-	System Architecture
UiTM	-	University Teknologi MARA
UTM	-	Universiti Teknologi Malaysia
XML	-	Extensible Markup Language

LIST OF APPENDICES

CHAPTER 1

INTRODUCTION

1.1 Overview

Universiti Tun Hussein Onn Malaysia (UTHM) is in the process of constructing a new on-premise data centre to facilitate the consolidation, modernisation, and oversight of its institutional IT infrastructure. Notwithstanding this strategic commitment, the institution does not own a formalised Enterprise Architecture (EA) that can deliver a cohesive framework aligning business processes, information systems, data assets, and foundational technical platforms. The lack of a formalised Enterprise Architecture has resulted in disjointed system implementations, segregated data repositories, restricted visibility of inter-system relationships, and considerable difficulties in achieving interoperability and effective data integration across many departments and functional areas. These conditions impair operational efficiency, restrict data-driven decision-making, and obstruct UTHM's capacity to attain sustainable digital transformation. In acknowledgement of these limitations, UTHM has demonstrated a willingness to accept external and academic contributions that can present a complete and adaptable EA model appropriate for institutional application.

This thesis addresses the institutional requirement by developing a comprehensive Enterprise Architecture for UTHM, informed by The Open Group Architecture Framework (TOGAF). TOGAF is utilised for its globally acknowledged structured methodology, focus on governance, and appropriateness for large, complex entities like higher education institutions. The proposed EA aims to create a cohesive and standardised architectural framework that aligns UTHM's strategic goals with its technological capacities, enhances integration among academic and administrative systems, improves governance and lifecycle management of IT assets, and facilitates scalable future digital initiatives. The study further creates a prototype enterprise system utilising SAP technology to operationalise and test the suggested design. This prototype illustrates the actual application of essential architectural elements, highlights the integration of enterprise resource processes within the EA framework, and assesses the viability of SAP as a fundamental platform to underpin UTHM's mission-critical operations.

This research seeks to furnish UTHM with a systematic, pragmatic, and future-oriented Enterprise Architecture model by integrating architectural design and system prototyping, thereby reinforcing institutional governance, fostering system interoperability, improving data integration, and advancing the university's enduring digital modernisation strategy within its new data centre environment.

1.2 Problem Statement

UTHM is in the process of establishing a new on-premise data center to support the centralization and modernization of its institutional IT infrastructure. UTHM currently does not possess a functional EA that clearly represents how its business processes, applications, data, and technology components are integrated and governed. This is because UTHM lacks EA expertise.

The lack of a well-defined EA has resulted in fragmented system management, limited visibility of system interdependencies, and difficulties in ensuring effective data integration. In order to solve this problem, UTHM has stated that it is open to external proposals, including academic contributions, that could demonstrate a suitable enterprise architecture model for potential adoption.

In response to this challenge, this case study focuses on improving the current system landscape in order to design an enhanced and functional Enterprise Architecture for UTHM. In addition, a prototype system utilizing SAP technologies is developed to allow UTHM to test an integrated enterprise system within the proposed architecture. This approach aims to demonstrate how SAP can be aligned with the EA to support core university operations, improve data integration, and validate the practicality of the proposed architecture within UTHM's new data center environment.

1.3 Project Objectives

- To design a comprehensive architectural framework that aligns the inventory system with UTHM’s organizational structure and ICT policies.
- To model and automate business processes that allow seamless handovers between departments.
- To establish a Single Source of Truth by designing an integrated enterprise data model and eliminating data redundancy and ensuring consistent asset information across all UTHM divisions.
- To develop a system capable of generating enterprise-level insights, helping UTHM management make data-driven decisions on budget planning and asset utilization.

1.4 Project Scope

The scope of this project is designed to fit in the project where this project focuses on providing a structured transition from UTHM's existing enterprise model to an integrated enterprise model based on their goals in digitalizing transformation. The main components of the project are:

1. Review existing system enterprise model

This project begins with an "As-Is" assessment of UTHM's current system environment based on the visit institutional observations. This phase focuses on analyzing how redundant processes and the lack of real-time data sharing hinder operational efficiency. The goal is to document the current relationship between business processes and evaluate how the current system lack of integration impacts institutional operational efficiency.

2. Design To-Be Enterprise Model

The project involves designing a comprehensive "To-Be" Enterprise Model that provides a scalable and integrated blueprint for the university that maps the four architectural layers (Business, Data, Application, and Technology) The scope includes a detailed design of Use Case Diagrams, Use Case Descriptions, Activity Diagrams, and Sequence Diagrams to model the behavioral logic of the subsystems. Additionally, it covers Database Design through Class Diagrams and Entity Relationship Diagrams (ERD) to ensure data integrity, and designing the interface to define the user experience and interaction for the individual subsystem within the new architecture.

3. System Development

This project also involves building a working prototype using the SAP Business Technology Platform to connect different departments and show how information flows smoothly. In this stage, the physical database is created to store all necessary data, and the main system functions are coded. This process demonstrates how an automated, integrated system makes university operations faster and more efficient.

1.5 Project Importance

This project is important as it helps UTHM address the current problems especially in UTHM's enterprise architecture at the same time to improve its operational efficiency, system integration and user experience. The project ensures that data flows smoothly across platforms like Learning Management System (LMS) and student portal by analyzing and enhancing the integration of subsystems such as student enrollment, course registration, hostel management and university healthcare. This improvement surely benefits students, staff and administrators by providing them with reliable access to information and services.

Optimising proposal enterprise architecture helps UTHM make better decisions, use resources more effectively, and ensure their IT systems can grow over time. This approach will take full advantage of new technologies like AI, IoT, and cloud computing to support their academic, administrative, and research work. In the end, this project strengthens UTHM's commitment to digital transformation, operational excellence, and ongoing improvement, helping them remain a modern and efficient university.

CHAPTER 2

LITERATURE REVIEW

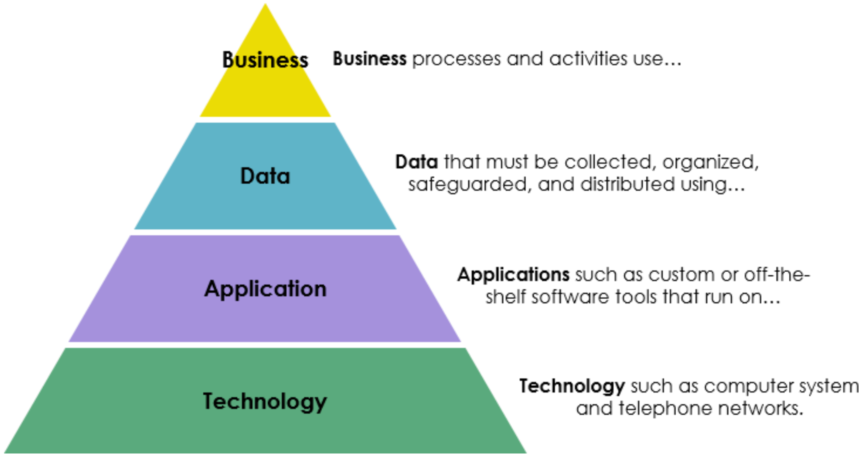
2.1 Introduction

This chapter provides a literature review of the key concepts that define the enterprise-level design and integration discussed in this project. The modern enterprise, especially in higher education, requires a comprehensive framework to strategically align its business objectives with its Information Technology (IT) capabilities and infrastructure. This framework is known as Enterprise Architecture (EA). We will also discuss the related sub-system that exist in universities EA by taking UiTM as an example. In the end, we are going to discuss technology used in EA which is the SAP Business Technology Platform and its details.

2.2 Enterprise Architecture (EA)

Enterprise Architecture (EA) is defined as a comprehensive framework designed to manage and align an organization's business objectives with its Information Technology (IT) capabilities and infrastructure (BOC Group, 2023; CIO Index, n.d.). It encompasses the systematic design, planning, and management of the organization's technology, processes, and systems to effectively support its strategic goals and objectives (BOC Group, 2023). EA provides a holistic view of the enterprise by organizing its structure into four interconnected architectural domains (Visual Paradigm, 2024)

Figure 1: *The Four Domains of Enterprise Architecture*



Note. Source: (Visual Paradigm, 2024).

Business Architecture: This domain maps the organization's strategy, governance, core business capabilities, value streams, and processes. Its central focus is on defining the "what" and "why" of the business operations, ensuring that technological investments are driven by strategic needs (CIO Index, n.d.).

Data Architecture: This domain establishes the principles and structures for how the organization's data assets are to be collected, stored, managed, and utilized. It is critical for ensuring data quality, consistency, and accessibility throughout the entire enterprise (BOC Group, 2023).

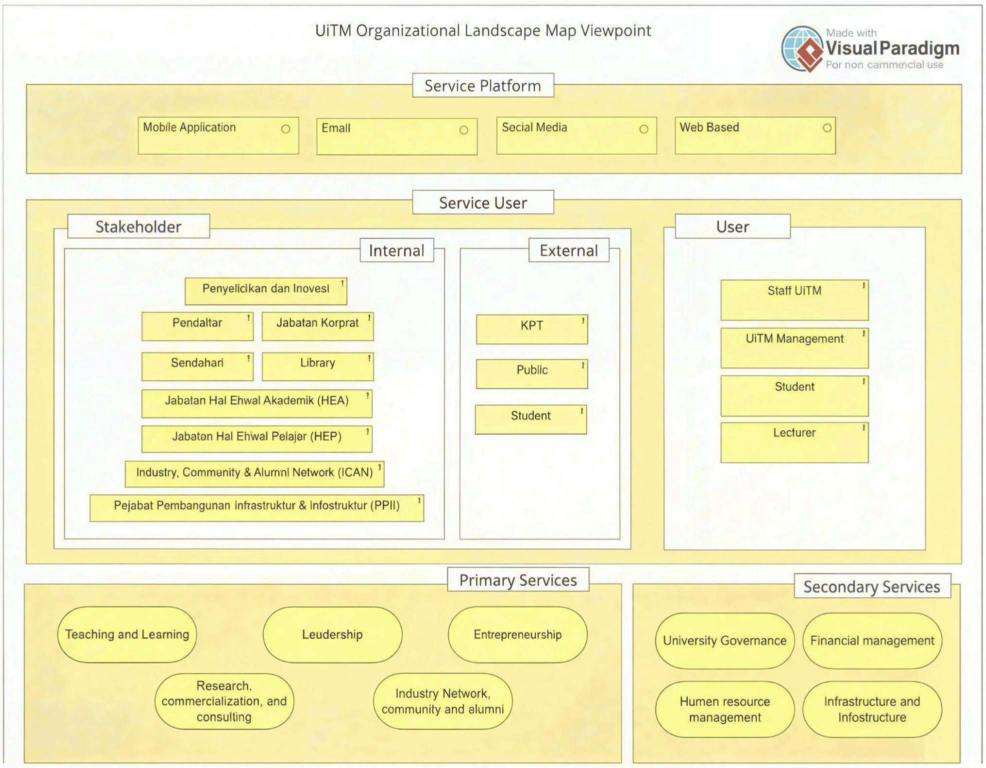
Application Architecture: This domain involves identifying the specific applications utilized within the enterprise and defining their interactions with each other and with external stakeholders. The objective is to establish an integrated, efficient, and scalable application portfolio (CIO Index, n.d.).

Technology Architecture: This domain describes the foundational hardware, software, networking infrastructure, and operating systems that collectively support the application and data layers. Its primary concern is ensuring the reliability, scalability, and security of the underlying technology foundation (BOC Group, 2023; CIO Index, n.d.).

2.3 Related Subsystem

Enterprise Architecture (EA) in higher education institutions typically integrates multiple subsystems that support academic, research, governance, administration, and community functions. These subsystems interact across different layers, including business processes, service delivery, information technology platforms, and stakeholder interactions, in an attempt to align institutions with their strategic goals. Many universities adopt EA for simplifying services provided to students and staff, standardising information management, and increasing decision-making effectiveness (Amin et al., 2024).

Figure 2: UiTM Organizational Landscape Map Viewpoint



Note: Source: (Amin et al., 2024).

Figure 2.2 shows UiTM's Enterprise Architecture (EA) landscape, which is built around different subsystems. Each subsystem handles specific tasks, such as academic services, administrative services and digital platforms. They all work together to connect stakeholders and service users. The framework includes internal and external stakeholders, main and supporting service areas and works across platforms like mobile, email, social media, and websites. By linking these subsystems, UiTM creates a smooth and integrated service environment that supports its goals for improving the institution (Amin et al., 2024).

Subsystem Category	Subsystem	Description
Service Platforms	Mobile App, Email, Social Media, Web-based Systems	Digital channels enabling communication and service access for users.
Stakeholders	Internal: HEA, HEP, Library, Corporate, ICAN, PPII, etc.	Organizational units involved in academic, student, infrastructure, alumni, and corporate services.
	External: KPT, Public, External Students	External bodies influencing or consuming UiTM services.
User Groups	UiTM Staff, Management, Students, Lecturers	Primary service users interacting with UiTM digital and administrative services.
Primary Services	Teaching & Learning, Leadership, Entrepreneurship, Research & Commercialization, Alumni & Industry Network	Core mission-driven services supporting UiTM's academic and developmental goals.
Secondary Services	University Governance, Human Resource, Finance, Infrastructure	Supporting subsystems enabling operations, compliance, and resource management.

Table 2.1 Description of UiTM EA Subsystems

Table 2.1 provides an overview of how each subsystem at UiTM supports the services offered at the university and the involvement of different stakeholders. It categorises subsystems into functional types, describes their purpose, and identifies service platforms on which users can access them. It also provides information on the internal and external stakeholders and user groups, and the primary and secondary services each subsystem enables. This structured view shows how UiTM integrates its digital and administrative systems into a seamless, mission-driven service environment.

Feature	Layered EA Model	UiTM Organizational Landscape Map Viewpoint
Main Focus	**Architectural Components** (What are the components of the enterprise's IT?).	**Organizational and Service Components** (Who uses the services and what services are offered?).
Business/Organizational Content	Lists **Functional Areas**: Academic Management, Finance, Human Resources, etc.	Lists **Key Stakeholders** (Internal/External), **Users** (Staff, Student, Lecturer), and **Services** (Teaching, Leadership, Governance).
IT/Technology Content	Lists **Abstract Domains**: Servers, Network, Middleware, Database (under Technology Architecture).	Lists **Delivery Channels/Platforms**: Mobile Application, Email, Social Media, Web Based (under Service Platform).
Framework Link	Strongly aligned with the **TOGAF/Zachman** frameworks' core layered structure.	Represents a **Viewpoint** or business-centric model, often part of an overall EA deliverable.

Table 2.2 Comparison between Layered EA Model and UiTM Organizational Landscape Map Viewpoint

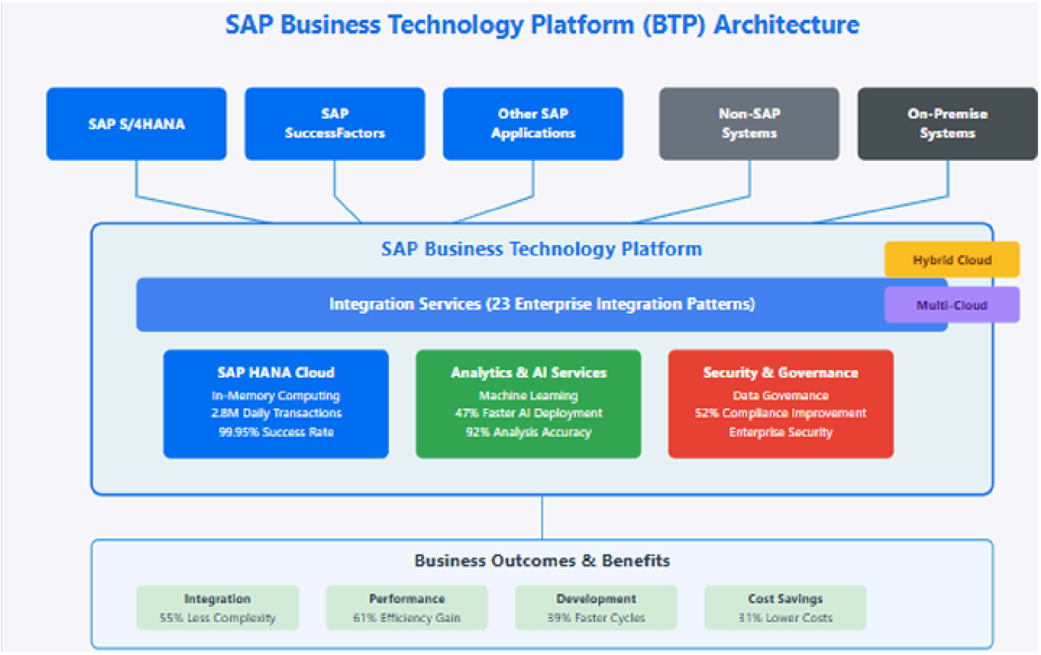
This Table 2.2 visual two representations of Enterprise Architecture (EA) serve distinct but complementary roles: the Layered EA Model acts as a structural blueprint, primarily focusing on Architectural Components by detailing the required Functional Areas like Academic Management and the underlying abstract IT Domains such as Servers and Network, consistent with the formal, tiered structure of frameworks like TOGAF; conversely, the UiTM Organizational Landscape Map Viewpoint functions as a high-level, business-centric communication tool, focusing on Organizational and Service Components by illustrating the key Stakeholders

(Staff, Students) and the Delivery Channels (Mobile Apps, Web) they use to access the university's services.

2.4 Technology Used

The SAP Business Technology Platform (BTP) is architecturally designed as a unified cloud foundation, acting as a crucial integration layer for the modern enterprise (Komarina & Sajja, 2025). It is built upon a framework that encompasses the Integration Suite, Extension Suite, and Data Management Suite, leveraging a microservices approach to provide a platform with defined scalability features and a structured security framework (Annanki, 2023).

Figure 4 : *SAP BTP Architecture*



Note: Source: (Komarina & Saja, 2025).

Core Components of SAP BTP

1. **Database and Data Management**
   Database and data management focuses on efficient data storage, processing, and governance. Key services include SAP HANA Cloud, SAP Data Warehouse Cloud, and SAP Data Intelligence, which collectively support real-time data access and enterprise-level data management.

2. **Analytics**
   SAP BTP includes analytical tools such as SAP Analytics Cloud (SAC), enabling organizations to perform reporting, data visualization, predictive analytics, and planning. These tools help businesses make informed, data-driven decisions.

3. **Application Development and Automation**
   This component offers capabilities for building and extending applications using low-code/no-code tools, cloud-native development frameworks, and process automation. Tools such as SAP Build Apps, SAP Build Process Automation, and the SAP Cloud Application Programming Model (CAP) streamline application development and workflow automation.

4. **Integration**
   SAP BTP supports seamless connectivity between SAP and non-SAP systems through services like SAP Integration Suite, API Management, and Event Mesh. These tools ensure smooth data exchange and interoperability across the organization's technology landscape.

SAP BTP enables organizations to modernize business processes, reduce dependency on custom code in core systems, and create flexible, scalable solutions. By unifying application development, data management, analytics, and integration services, the platform helps businesses enhance innovation, improve efficiency, and adapt quickly to changing operational needs.

2.5 Summary

This chapter establishes the theoretical foundation for the project by exploring Enterprise Architecture (EA) as a tool to align university business goals with IT infrastructure. By examining the four key domains which are Business, Data, Application, and Technology, the review clarifies how a system must be structured to support an entire organization.

Through the case study of UiTM’s EA landscape, the chapter illustrates how various subsystems and stakeholders interact to create an integrated service environment. Finally, the review of SAP Business Technology Platform (BTP) identifies the modern technology needed to achieve this integration, focusing on data management, analytics, and seamless system connectivity. Together, these elements provide the blueprint for designing the UTHM Inventory System as a robust enterprise solution.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter is a description of the methodology of Waterfall with System Development Life Cycle (SDLC) used to help with the planning, analysis, design and implementation of improvements to UTHM's Enterprise Architecture framework. The Waterfall of SDLC uses formal, creating or altering systems, and the models and methodologies that people use to develop these systems (Kute & Thorat, 2014). The choice to use a structured methodology focused on requirements, architectures and specifications being fully defined and validated earlier in the process was due to the increased difficulty in changing or correcting Enterprise Architecture projects after they have been implemented due to their high cost of implementation and complexity to implement from start to finish. The following chapters explain each phase of the methodology, including phases, activities and deliverables, provide a project schedule like for example using a Gantt Chart or Kanban format with major milestones, and summarise the entire development of the project.

3.2 Proposed Methodology

This study used the Waterfall System Development Life Cycle (SDLC) model as its approach. The Waterfall methodology is chosen for its systematic, documentation-centric, and linear characteristics, making it appropriate for major business projects like the creation of business Architecture (EA) for Universiti Tun Hussein Onn Malaysia (UTHM). Considering that EA projects entail substantial implementation costs, widespread institutional effects, and long-term strategic coherence, a methodology prioritising meticulous planning, requirement verification, architectural delineation, and regulated execution is necessary. The Waterfall SDLC facilitates thorough analysis and robust architectural design prior to deployment, reducing rework and maintaining architectural consistency throughout the project lifecycle.

This study uses the Waterfall technique to direct the design and development of an improved Enterprise Architecture framework for UTHM, alongside the creation of an enterprise-wide SAP prototype to validate the proposed architecture. Each phase of the technique concentrates on systematically converting UTHM's institutional requirements into a structured design, subsequently validated through technical implementation. Stakeholder interaction is integrated to guarantee conformity with institutional expectations and operational reality. The suggested methodology has five primary phases: Feasibility Study, System Analysis, System Design, Implementation, and Maintenance and Support.

3.3 Phases of Methodology

The project follows the five core phases of the SDLC which is Feasibility Study, System Analysis, System Design, Implementation and Maintenance and support

Phase 1:Feasibility Study

This first stage will determine whether the project is viable. We adhere to UTHM's standard operating procedure for new system requests.

- FSR Process: The project began with the submission of the Feasibility Study Report (FSR).
- Committee Approval: FSR is submitted to the IT Security & Steering Committee which will evaluate the proposal.

Outcome: The "Go /No-Go" decision based on the project proposal and literature review.

Phase 2: System Analysis

In this phase , we collect the detailed business requirements to bridge the gap between the definition of the requirement and how it should be.

- Data Collection: Conducting interviews with UTHM stakeholders (Asset Unit officers) to understand the current manual form and workflow.
- Requirement Specification: Defining the user roles and their access privileges.

Phase 3: System Design

In this phase , we convert the requirements into technical specifications.

- Architecture Design: We utilize the three-tier architecture model to ensure the system is scalable and easy to maintain.
- Database Design:Create the entity-relationship diagram to model the relationships between asset, staff and location.
- Interface Design: Prototyping the dashboard UI to ensure they are user-friendly for non technical staff.
- Infrastructure Design: The system will be hosted in UTHM's Data Center. It will benefit from high-availability infrastructure that includes dual power sources, UPS backup and generator set to ensure very high uptime.

Phase 4: Implementation

This is the phase where the system is built and implemented.

- Coding: Developing the subsystem using web technologies such as PHP and java then connect it to the database MySql.
- Data Migration: Adopt a phased migration strategy, which converts sample data from existing Excel sheets into the new system format.
- Security Implementation: The code will be hosted behind the UTHM's existing security layer which include 2 Firewalls (Europe/China) and a Web Application Firewall to protect the portal from being attacked from the external side.
- Testing: Perform testing on each individual module and integration testing to ensure it communicates with the Financial module.

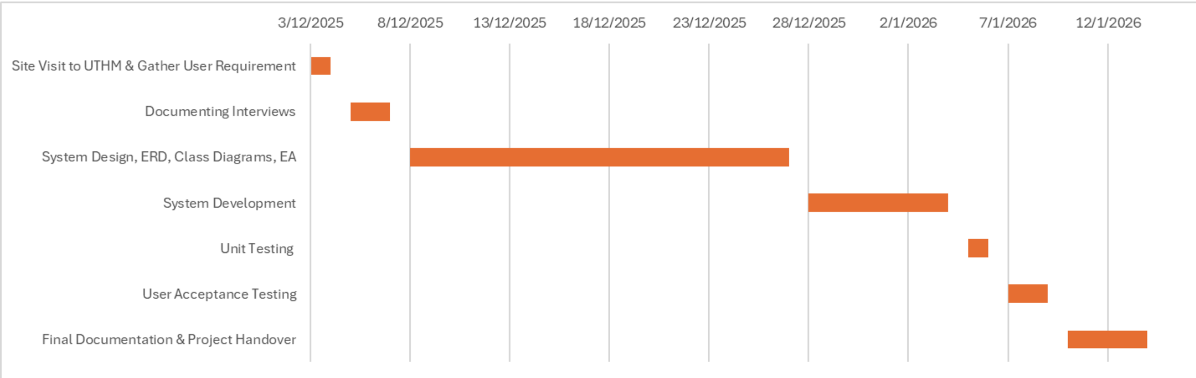
Phase 5: Maintenance and Support

In this phase , we involve training and patching activities. We will teach the UTHM staff how to use the new QR code features and fix any bugs that are discovered after the system is deployed.

3.4 Project Planning Schedule

In order to make sure this project runs smoothly, a schedule consists of every phase and its duration is made. The team members that are in charge of leading the task are also determined to make sure all the phases are supervised.

Figure x: Project Schedule Gantt Chart



Tasks	Start Date	End Date	Task Leader
Site Visit to UTHM & Gather User Requirement	3/12/2025	4/12/2025	Aisyah
Documenting Interviews	5/12/2025	7/12/2025	Harini
System Design, ERD, Class Diagrams, EA	8/12/2025	27/12/2025	Zhiming
System Development	28/12/2025	4/1/2026	Afiq
Unit Testing	5/1/2026	6/1/2026	Auni
User Acceptance Testing	7/1/2026	9/1/2026	Ziyaad
Final Documentation & Project Handover	10/1/2026	14/1/2026	Ika

3.5 Summary

This chapter has detailed the Waterfall System Development Life Cycle (SDLC) methodology chosen to guide the improvement of UTHM’s Enterprise Architecture. By utilizing the five core phases which are feasibility study, system analysis, system design, implementation, and maintenance, the project minimizes the risks associated with the high cost and complexity of enterprise-level systems.

This structured path ensures that the resulting SAP prototype is not only technically sound but also strategically aligned with UTHM's specific operational needs and secure within its new data center environment. The inclusion of a detailed project schedule and Gantt chart ensures that all tasks, from the initial site visit to system implementation and testing are supervised and completed within the established timeframe. Ultimately, this structured methodology serves as the bridge between UTHM’s strategic digital transformation goals and the successful delivery of a secure, integrated, and high-performance enterprise environment.

CHAPTER 4

Analysis and Design

4.1 Introduction

This chapter analyses the organisational structure of Universiti Tun Hussein Onn Malaysia (UTHM), the current system environment, and enterprise requirements to facilitate the development of a complete Enterprise Architecture (EA) framework and a SAP-based enterprise prototype. A comprehensive comprehension of the university's organisational structure, operational processes, and information dissemination methods is crucial for ensuring that the suggested design is congruent with institutional goals and effectively tackles existing operational issues. Given that UTHM presently functions with several autonomous systems exhibiting minimal integration, this analysis establishes the essential groundwork for delineating how a cohesive and centralised enterprise framework might improve institutional efficiency, governance, and data-informed decision-making.

The chapter initially analyses UTHM’s organisational framework at the Strategic, Tactical, and Operational tiers to pinpoint principal decision-makers, stakeholders, and primary data generators. This is succeeded by a comprehensive evaluation of the existing system landscape, emphasising the dependence on disjointed digital platforms and semi-manual procedures, although the presence of a contemporary three-tier data centre architecture. A comparative analysis of the current and proposed environments is subsequently provided to demonstrate the shift from isolated systems to a fully integrated enterprise ecosystem, facilitated by centralised data management, Single Sign-On (SSO), and an improved user experience.

To guarantee that the suggested architecture appropriately represents institutional demands, requirements gathering activities were performed through stakeholder involvement via structured interviews. These interactions facilitated the recognition of essential business requirements, operational obstacles, technical anticipations, and governance factors. The chapter finishes by delineating the functional and non-functional requirements that will inform the design, development, and deployment of the improved EA framework. These investigations collectively establish the analytical foundation for the ensuing architectural design and solution implementation phases.

4.2 Company Organization Structure

A successful system integration requires clear understanding of the organizational hierarchy and the information flow between stakeholders. UTHM operates on three distinct levels which are Strategic, Tactical and Operational. This structure must be bridged to eliminate the "Vertical Silos" which are often found in large organizations.

1. Strategic Level
    - IT Security and Steering Committee: This high-level body responsible for the overall digital strategy for UTHM. They will review and approve the Feasibility Study Reports for all proposed systems. They need to determine whether the new system is necessary and aligns with the university budget and culture.
    - Management Implications: Top management support is critical to enforce the cultural changes to adapt the new system.
2. Tactical Level
    - Bursary Department (Bendahari) : Bursary defines the rules that the software must obey. They act like a central hub for data verification, requiring inputs from various faculty to ensure accuracy.
3. Operational Level
    - Faculties and Responsibility Centers: This level includes academic and administrative staff that are distributed across the campus. They are the primary data generators.

4.3 Current System Analysis

UTHM relies on a range of digital systems and technologies to support its academic, administrative, and operational activities. Online platforms are used for teaching and learning, student administration, finance, human resource management, and communication between students and staff. These systems enable basic information access and daily operations. However, they are implemented independently to meet specific departmental needs. As a result, data sharing across departments are limited and operational efficiency depends on manual processes. In terms of infrastructure, UTHM is in the process of developing a new three-tier data center to host and support its digital systems. While this infrastructure provides a strong physical foundation for digital services, its capabilities have not yet been fully utilized in the current system environment as there are no centralized system management and integrated digital solutions. This situation highlights the need for improved system coordination and structured planning to maximize the benefits of the new data center.

4.4 Comparison between Existing and Proposed System

The transition from UTHM's current digital landscape to the proposed architecture represents a shift from a fragmented system to a unified digital system. Currently, the university relies on independent systems that serve specific departmental needs. While a modern three-tier data center exists, its potential remains not fully utilized because the current systems lack the centralized management and integration needed to utilize such a foundation. The proposed system addresses these limitations by replacing manual workarounds with end-to-end digitalization, guided by a clear "Digital Transformation" vision. Instead of isolated data, the proposed architecture introduces a centralized data warehouse and automated pipelines to ensure information flows seamlessly across the institution. This change moves towards a more agile, hybrid cloud environment. Furthermore, the user experience is improved from using multiple disconnected platforms to a streamlined approach featuring Single Sign-On (SSO) and a unified mobile app.

4.5 System Requirements Gathering Technique

To ensure the proposed UTHM Inventory System aligns with the actual operational needs of the university, the team employed the Interview Method as the primary data collection technique. This qualitative approach allowed the team to explore the complexities of existing workflows and identify specific pain points across different hierarchical levels. The interview method was selected due to its flexibility and the depth of information it provides. Compared to other methods like questionnaires, interviews allow the team to capture the specific needs of both the administrative staff and the end-users. The interviews were conducted during the site visit on 3/12/2025. The team roles were distributed as follows:

- **Lead Interviewer (Aisyah):** Facilitated the discussion using a prepared set of open-ended questions.
- **Technical Observers (Afiq & Ziyaad):** Focused on identifying technical constraints and integration points with current UTHM legacy systems.
- **Note-Takers (Ika & Harini):** Documented the responses and captured non-verbal cues regarding system frustrations.

Below are the sample of questions interviewed and its answers:

Q: Are there still manual methods or all is digitalized?

A: UTHM is not 100% digital yet. One of the departments that are still running manually are their Health Center. However, the admin has been enforcing the lecturer to use LMS and work on a singular system.

Q: Is there a stand alone database different from the main database?

A: Only LMS has separated database but all the system get data in on single database

Q: Who approves of the new system?

A: The team asks for user requirements first, then the team will develop the system and send it to the committee to decide either to accept or reject the new system.

4.6 System Requirements

Functional Requirements

Functional requirements specify the particular functionality and features that an updated enterprise architecture (EA) framework should offer to UTHM to help streamline primary business operations in UTHM's IT ecosystem. Functional requirements help to ensure that the updated EA framework provides support for key UTHM processes such as asset management, decision making and compliance.

- User Authentication - A secure account accessible for all UTHM staff and students using Single Sign-On (SSO).
- Requirements Management - All systems requirements should be captured, traced and validated from the SDLC analysis phase and then linked to the design output.
- Reporting - Generate customizable reports that should be exportable along with Gantt charts and phase deliverables that can be viewed by administrators.
- Integration Support - EA Framework must provide the ability to import and export data that is in existing tools like TOGAF during integration activities.

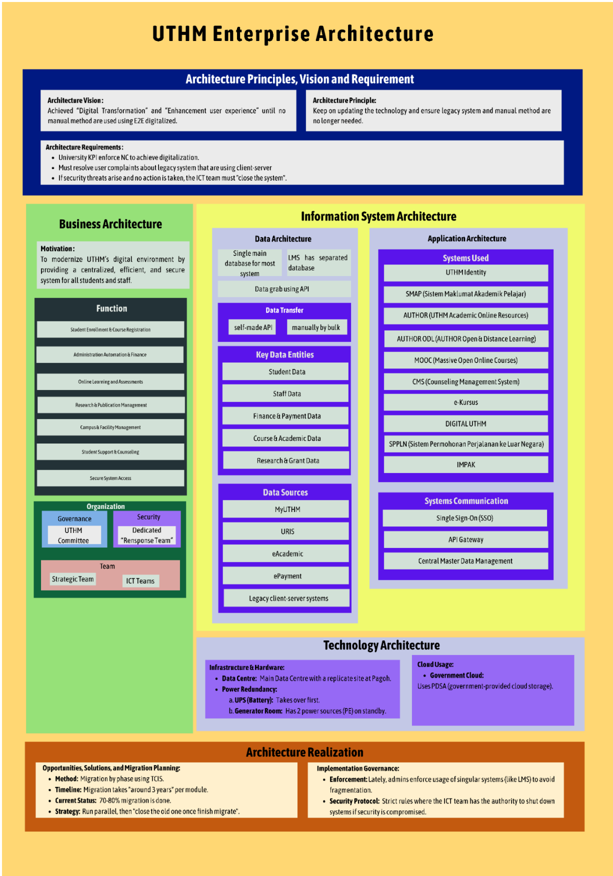
Non-Functional Requirements

Non-functional requirements specify the quality attributes and performance standards the EA framework must meet, ensuring reliability, efficiency, and usability in UTHM's demanding academic environment.

- Performance: System shall load web pages within 3 seconds for datasets up to 1,000 components.
- Scalability: System must support around 2000 concurrent users.
- Security: Encrypt data especially for passwords and using firewalls like Web Application Firewall (WAF) on the webpage system.
- Usability: Intuitive for new users and accessible for various users.
- Reliability: The system must have 99.5% uptime.

4.7 System Design

4.7.1 Enterprise Architecture AS-IS



4.7.2 System Design AS-IS

This section explains the current architecture at UTHM before the new system is implemented.

1. Business Architecture
   - **Strategy:** Guided by "Digital Transformation 2030" to shift the culture from manual to digital, but administrative tasks still require manual forms currently.

- **Governance**: All proposed systems must pass the IT Steering Committee via a Feasibility Study Report and approved based on necessity and department expertise.
- **Operations**: Departments operate in horizontal silos. The LMS and TCIS are currently run as separate business entities with disconnected workflow.

2. **Application Architecture**
   - **Core system**: Legacy TCIS based on client-server technology.
   - **Accessibility**: Restricted on campus LAN. Staff cannot access the system while outstation without a VPN , this leads to bad user experience.
   - **Integration**: Applications are fragmented. For example the LMS is a separate application from the main student database, requiring API to bridge the gap.

3. **Data Architecture**
   - **Storage**: Data is distributed and decentralized. For example the student marks and physical asset record are separate, siloed databases.
   - **Synchronization**: The data transfer relies on API calls or need manual entry , rather than real-time synchronization.
   - **Performance**: Current database structure struggles with high concurrency or usage, causing up to 5- minute lags during peak usage.

4. **Technology Architecture**
   - **Facility**: Tier 3 Data Center located on the upper floor (flood prevention).
   - **Server Hosting**: Uses Hot Aisle/Cold Aisle Containment for thermal efficiency.
   - **Power**: High availability via Dual Power Sources (TN A/TN B), backed by UPS and Generator Sets.
   - **Security**: Perimeter protection using 2 Firewalls (Europe/China brands) and a Web Application Firewall (WAF) for portal defense.

4.7.3 Enterprise Architecture To-Be

Figure 4.2 illustrates the Enterprise Architecture To-Be for UTHM. It represents the future system structure of the university. This architecture helps improve system efficiency and supports UTHM's future digital development.

TO-BE

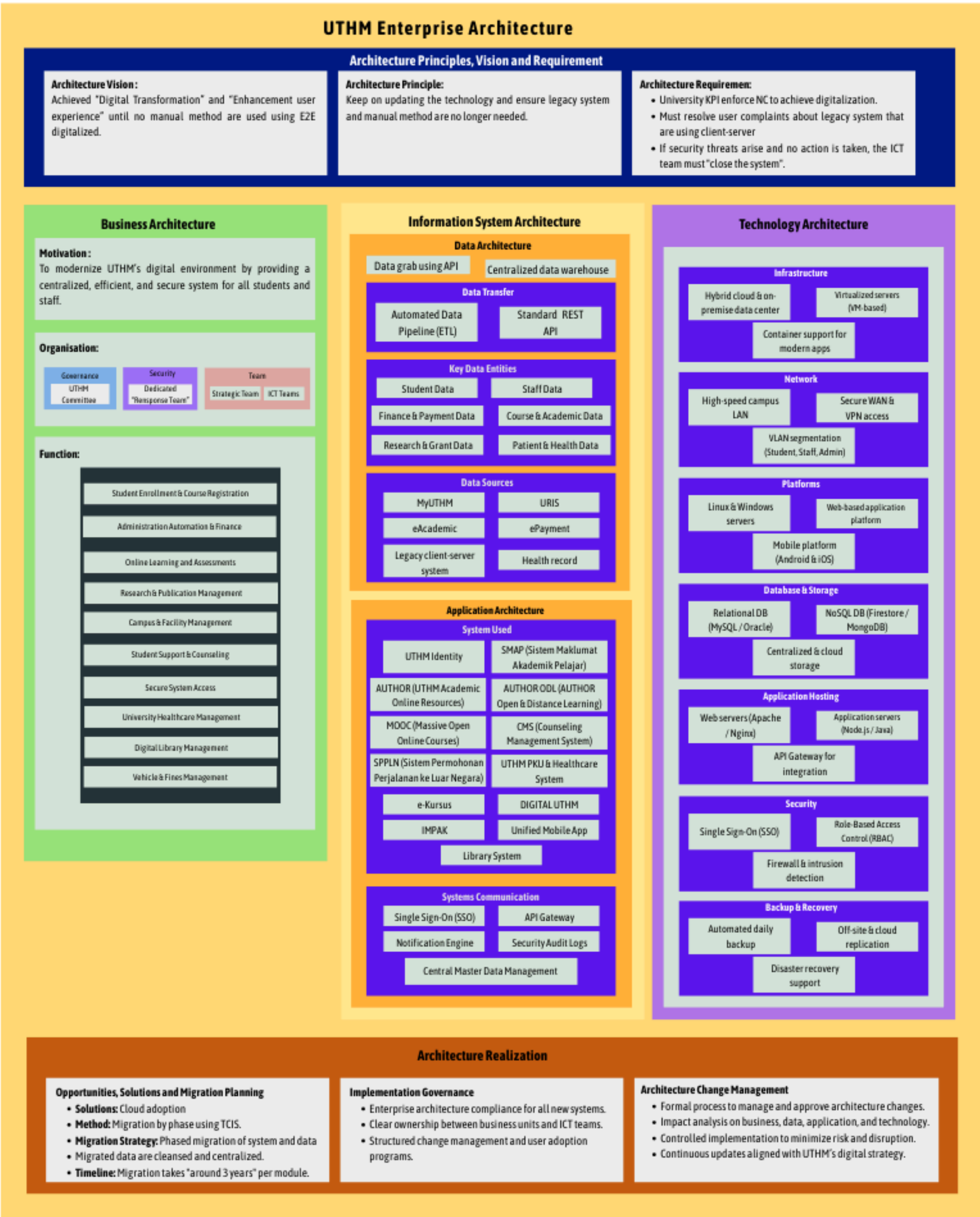


Figure 4.2 UTHM Enterprise Architecture To-Be

4.7.4 To-Be

This section explains the Enterprise Architecture To-Be as shown in Figure 4.2. The framework is organized into several main layers using TOGAF (The Open Group Architecture Framework). Each layer serves a specific purpose.

1. Architecture Vision and Principles

The top layer establishes the foundation for the EA. It comprises:

- Architecture Vision: Defines UTHM's goal of achieving digital transformation and improved telecommunication user experience until minimal manual processes.
- Architecture Principles and Requirements: Focus on updating technology to replace legacy systems until no manual method is needed.

2. Business Architecture

This layer, shown in green, describes what the university does and how it operates:

- Motivation: To develop a centralized, secure and efficient digital environment accessible to students and staff on any device at any time.
- Organization: Illustrates UTHM's departmental hierarchy.
- Functions: The architecture supports ten core business functions:
    1. Student Enrollment & Course Registration: Manages student admissions, course selection, and registration processes in a streamlined manner.
    2. Administration Automation and Finances: Manages administrative tasks, payroll, payment processing and allocation of university resources.
    3. Online Learning and Assessment: Support student learning and coursework using online platforms.
    4. Research and Publication Management: Supports research project management, grant applications and innovation initiatives.
    5. Campus and Facility Management: Coordinates maintenance, space allocation, and utilization of campus facilities.

6. Student Support and Counselling: Manages student welfare services, counselling sessions, and academic support programs.
7. Secure System Access: Authorized access to systems through identity management and authentication mechanisms.
8. University Healthcare Management: Supports the management of healthcare services and medical records within the university.
9. Digital Library Management: Provides access to digital academic resources, journals and library services.
10. Vehicle and Fines Management: Manages vehicle registration, parking systems and fine enforcement processes.

3. Information System Architecture

Divided into Data Architecture and Application Architecture:

- Data Architecture: Includes a centralized data warehouse that consolidates data from multiple systems. It supports dashboards for data visualization, automated data pipeline and data protection mechanisms. REST APIs are used to enable secure and standardized data exchange between systems.
- Application Architecture: The Application Architecture consists of multiple application layers that support academic, administrative, and operational activities at UTHM.
    1. Student & Academic Systems: UTHM Identity, SMPA, MyUTHM Academic, Open and Distance Learning (ODL) platform, Massive Open Online Courses (MOOCs), Author ID, and online academic resources.
    2. Administrative & Support Systems: Counselling Management System (CNS), SPLPN, UTHM Healthcare System, e-Kursus, DIGITAL UTHM, IMPAK, and Vehicle and Facilities Management systems.
    3. Library & Access Systems: Library System and Unified Mobile Application, providing centralized access to academic and student services.
    4. Integration: Connects all systems through Single Sign-On (SSO), API Gateway, notification engine, security audit logs, and centralized master data management

4. Technology Architecture

Defines the infrastructure supporting all systems and is organized into seven domains:

- Infrastructure: Hybrid cloud access and on-premise data center with virtualization support to ensure scalability, flexibility and efficient resource utilization.
- Network: High-speed backbone network, secure Wan and VPN access to support secure and advanced connectivity across the campus.
- Platform: Linux and Windows servers and web and mobile applications to support the deployment and execution of modern applications..
- Database: Relational (MySQL/Oracle), NoSQL (Firebase/MongoDB) and cloud storage for high availability.
- Application Hosting: Web servers such as Apache and Nginx, cloud-based hosting services, and API Gateway to host and integrate university applications.
- Security: Single Sign-On (SSO) , secure internal communications, firewalls, and intrusion detection to protect applications, data, and network resources.
- Backup: Automated backups, offsite/cloud replication, and disaster-recovery support data protection and business continuity.

5. Architecture Realization

This layer outlines implementation strategies:

- Opportunities, Solutions, and Migration Planning: Focus on cloud adoption and phased migration, prioritizing critical systems with integrated data preprocessing over a 3-year roadmap.
- Implementation Governance: Maintains alignment between EA and ICT teams, enforces compliance, and manages change programs.
- Architecture Change Management: Formalizes change approval, engages stakeholders across domains, minimizes risks and provides ongoing updates aligned with UTHM’s digital strategy.

4.7.5 System Architecture

Figure 4.3 below illustrates the system architecture of the UTHM Inventory Subsystem. Users access the system through the Web Interface, and all requests are transmitted via the Internet. Users are first authenticated using UTHM Single Sign-On (SSO) before any system functionality can be accessed. Once authenticated, requests are processed by the API Layer, which routes them to the appropriate application modules. These modules execute the required business logic and interact with the Inventory Database to retrieve or update data. The processed results are then returned through the API Layer to the frontend for display.

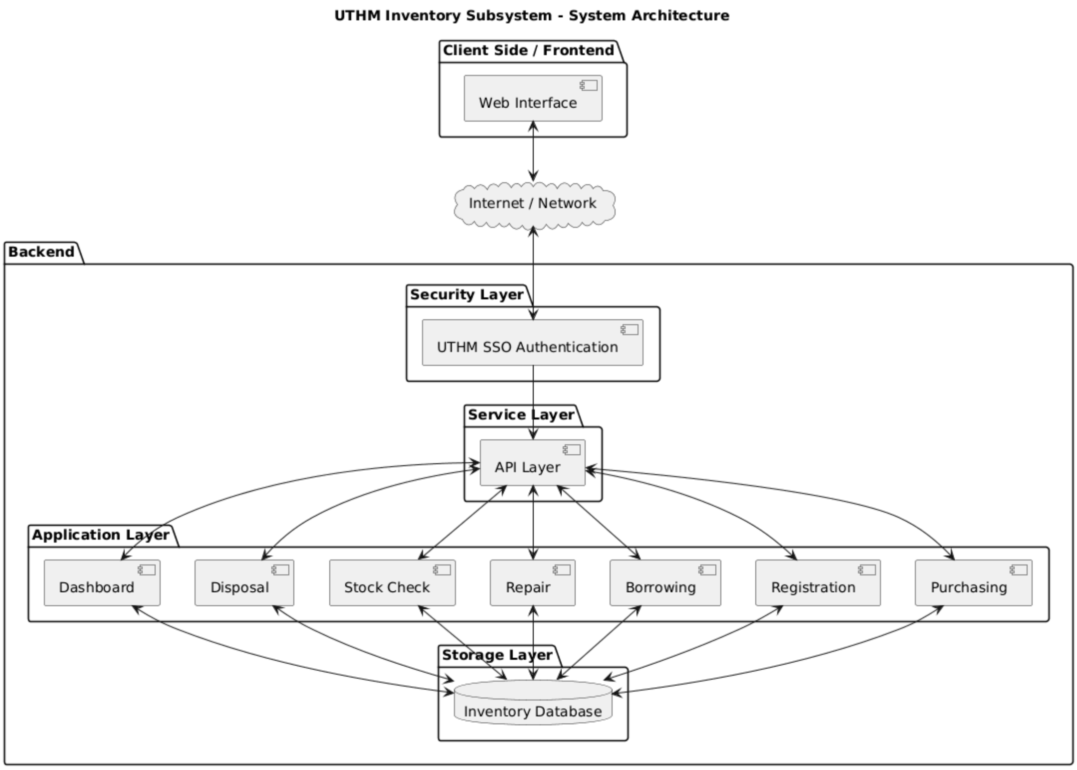


Figure 4.3 UTHM Inventory System Architecture

4.7.6 System Architecture Components

The UTHM Inventory Subsystem is designed using a cloud architecture. As shown in Figure 4.3, it divides the system into presentation which is the frontend, security, service, application and data layers. This design has a clear separation of responsibilities, maintainability and secure access to inventory data. The following explains the components in each layer and their functions.

1. Presentation Layer (Client Side / Frontend)
   - Component: Web Interface
   - Function: The presentation layer is the user-facing part of the system, where staff and students interact with the inventory subsystem. It handles input forms, dashboards and reports. It also presents all the data and insights in an intuitive and readable format. The frontend communicates with the backend over the network but does not perform any business logic or data storage operations.

2. Transport Layer (Internet / Network)
   - Component: Internet or Network Cloud
   - Function: This layer provides the connectivity between the frontend and backend. It enables secure transmission of requests and responses and allows users to access the system remotely while protecting data and confidentiality during transit.

3. Security Layer
   - Component: UTHM SSO Authentication
   - Function: Security is the first point of access for any system request. The UTHM Single Sign-On (SSO) verifies user credentials and enforces access control to ensure that only authorised personnel can perform operations such as purchasing, borrowing or stock checking. Authentication occurs before any business logic is executed to provide a secure foundation for the system.

4. Service Layer (API Layer)
   - Component: API Layer
   - Function: The service layer acts as a gateway between the frontend and the application modules. It receives authenticated requests from the user interface, validates requests and routes them to the appropriate business logic modules and returns processed results to the frontend. This separation ensures

that the frontend does not directly access backend modules or databases. This way, security risks are reduced and modularity is achieved.

5. Application Layer (Business Logic Layer)

- Components: Purchasing, Registration, Borrowing, Repair, Stock Check, Disposal, Dashboard
- Function: The application layer contains the core business logic of the inventory system. It consists of eight functions:

    1. Purchasing: To add new inventory items.
    2. Registration: To manage registration of users or items.
    3. Borrowing: To track check-outs and returns.
    4. Repair: To handle repair requests for damaged items.
    5. Stock Check: To provide real-time inventory status and reports.
    6. Disposal: To manage retirement of obsolete items.
    7. Dashboard: To summarize key metrics for management.

    All modules interact with the data layer to read or update inventory records that are being made by the administrator, lecturers or students of UTHM.

6. Data Layer (Storage Layer)

- Component: Inventory Database
- Function: The data layer stores all persistent system data including inventory records, user information and transaction history. It establishes data consistency, integrity and availability for all application modules.

4.7.7 Project Design

In this individual project design, disposal items are selected from the existing inventory system once they are identified as damaged or no longer usable. The selected items go through a disposal request, approval, and recording process to ensure proper control, accountability, and documentation.

Use Case Diagram

Figure 4.4 illustrates the use case diagram of the Disposal module within the Inventory System. It shows how different actors interact with the system to manage the disposal of inventory items. The main actors involved are Inventory Staff, Supervisor, and Disposal Committee.

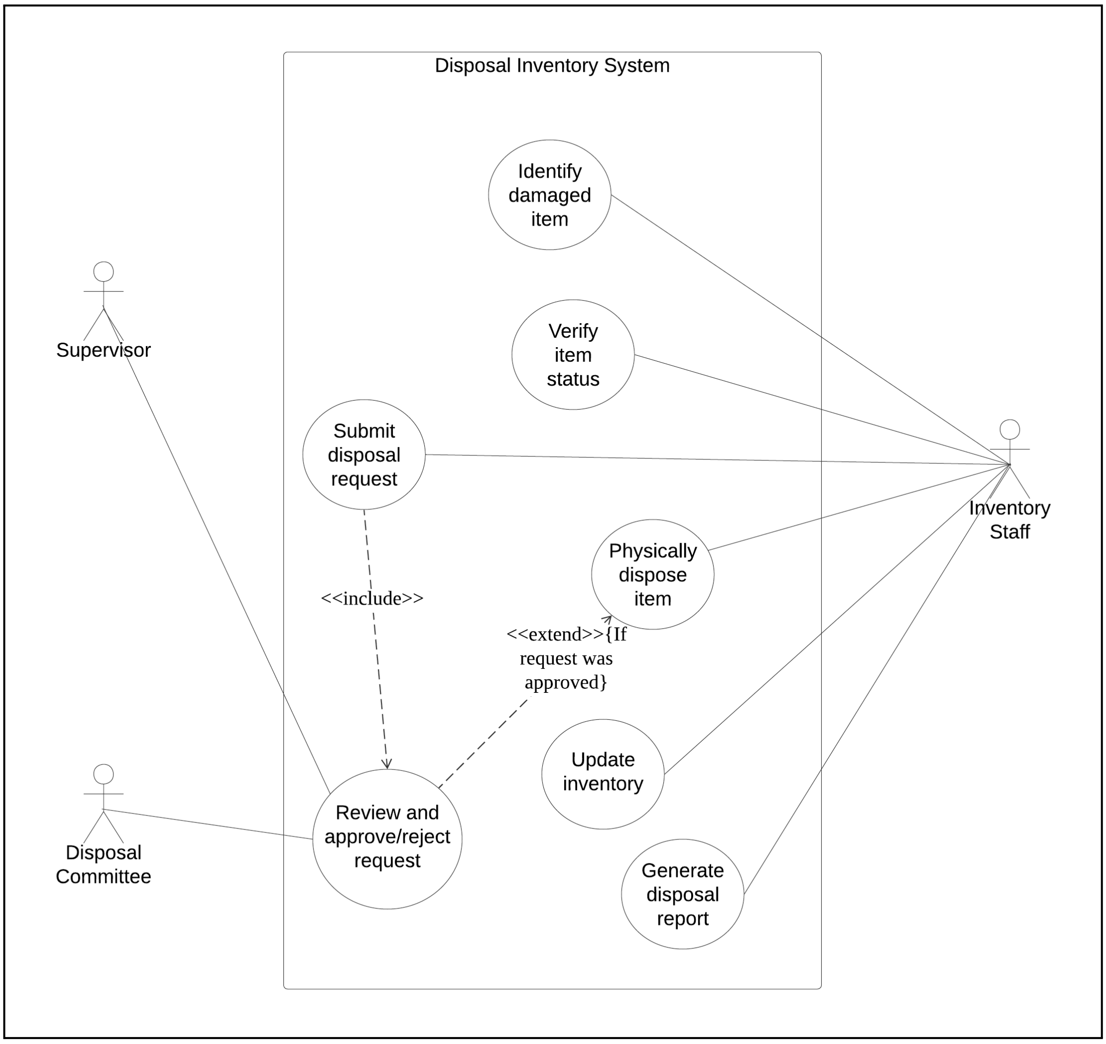


Figure 4.4 Use Case Diagram of Disposal Inventory System

Use Case Description

Use Case: Identify Damaged Item
ID: UC-D1
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database
Flow of events: 1. The use case starts when Inventory Staff noticed or informed that the item has been damaged. 2. Inventory Staff searches for the item in the system using item ID. 3. The system displays the item details. 4. Inventory Staff inspect the condition of the item. 5. Inventory Staff updates the item with a label as "Damaged". 6. Inventory Staff records damaged details. 7. The system saved the updated status and damages details.
Postcondition: 1. The item is recorded in the system as "Damaged" and becomes unusable from inventory usage.

Use Case: Verify Item Status
ID: UC-D2
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database
Flow of events: 1. The use case starts when Inventory Staff needs to confirm the current status of the item. 2. Inventory Staff searches for an item using item ID. 3. The system retrieves and displays the item details including current status. 4. Inventory Staff reviews the displayed status and related history.
Postcondition: 1. Inventory Staff has confirmed the item's latest status of the system.

Use Case: Submit Disposal Request
ID: UC-D3
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and has status “Damaged” which is eligible for disposal.
Flow of events: 1. The use case starts when Inventory Staff decides that a damaged item should be disposed of. 2. Inventory Staff opens the “Disposal Request” page. 3. Inventory Staff select the item from the inventory list by item ID. 4. Inventory Staff fills in disposal details or reasons. 5. Inventory Staff submit the disposal request. 6. The system forwards the request to the Supervisor/Disposal Committee for review.
Postcondition: 1. A disposal request is stored in the system with status “Pending Approval”.

Use Case: Review and Approve or Reject Request
ID: UC-D4
Actor: Supervisor and Disposal Committee
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and there is at least one disposal request with status “Pending Approval”.
Flow of events: 1. The use case starts when the Supervisor decides to review pending disposal requests. 2. The system displays a list of pending requests. 3. Supervisor/Disposal Committee selects a specific request to review. 4. The system shows requested item details, damage information, and justification for disposal. 5. Supervisor/Disposal Committee decides to approve or reject the request. 6. The system updates the request status to “Approved” or “Rejected”.
Postcondition: 1. The disposal request has a final decision status “Approved” or “Rejected” recorded in the system.

Use Case: Physically Dispose Item
ID: UC-D5
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and has a disposal request for the item that has status “Approved”.
Flow of events: 1. The use case starts when Inventory Staff needs to carry out the approved disposal. 2. Inventory Staff reviews the approved disposal request and instructions. 3. Inventory Staff disposes the item according to the approved method and university policy. 4. Inventory Staff records disposal details in the system. 5. The system saves the disposal record.
Postcondition: 1. The item has been physically disposed of and the disposal action is recorded in the database of the system.

Use Case: Update Inventory
ID: UC-D6
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database. 3. The item has been physically disposed of or its status has changed due to disposal processing.
Flow of events: 1. The use case starts when Inventory Staff needs to reflect back disposal in the inventory records. 2. Inventory Staff searches for the disposed item by ID. 3. The system displays the item details and disposal record. 4. Inventory Staff updates the inventory status to “Disposed”. 5. The system updates stock counts and asset lists accordingly.
Postcondition: 1. The disposed item no longer appears in active inventory or its database.

Use Case: Generate Disposal Report
ID: UC-D7
Actor: Inventory Staff
Preconditions: 1. The inventory system already been logged in using SSO 2. The item exists in the inventory database and contains disposal records.
Flow of events: 1. The use case starts when Inventory Staff needs a report of disposal activities. 2. Inventory Staff opens the "Disposal Report" application in the system. 3. Inventory Staff confirms the selected filters and requests to generate the report. 4. The system retrieves matching disposal records from the database. 5. The system generates the disposal report in the selected format.
Postcondition: 1. A disposal report is generated and available to Inventory Staff for review, download, or printing.

Activity Diagram and Sequence Diagram

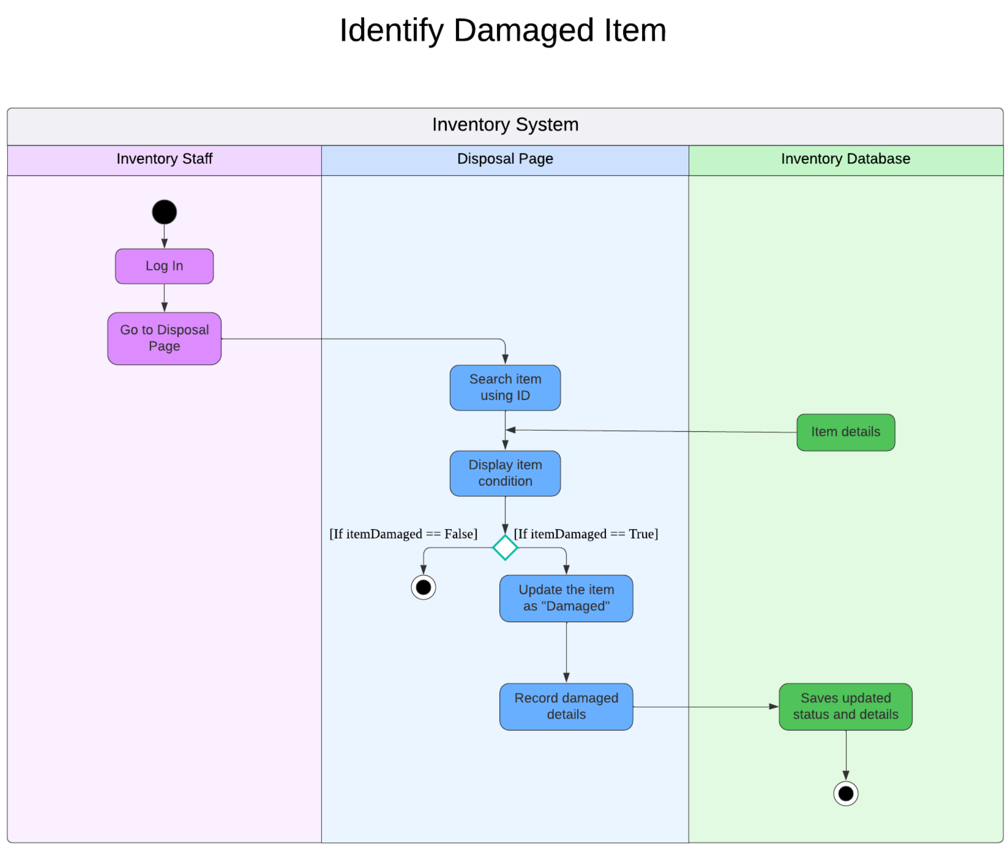


Figure 4.5 Activity Diagram of Identify Damaged Item

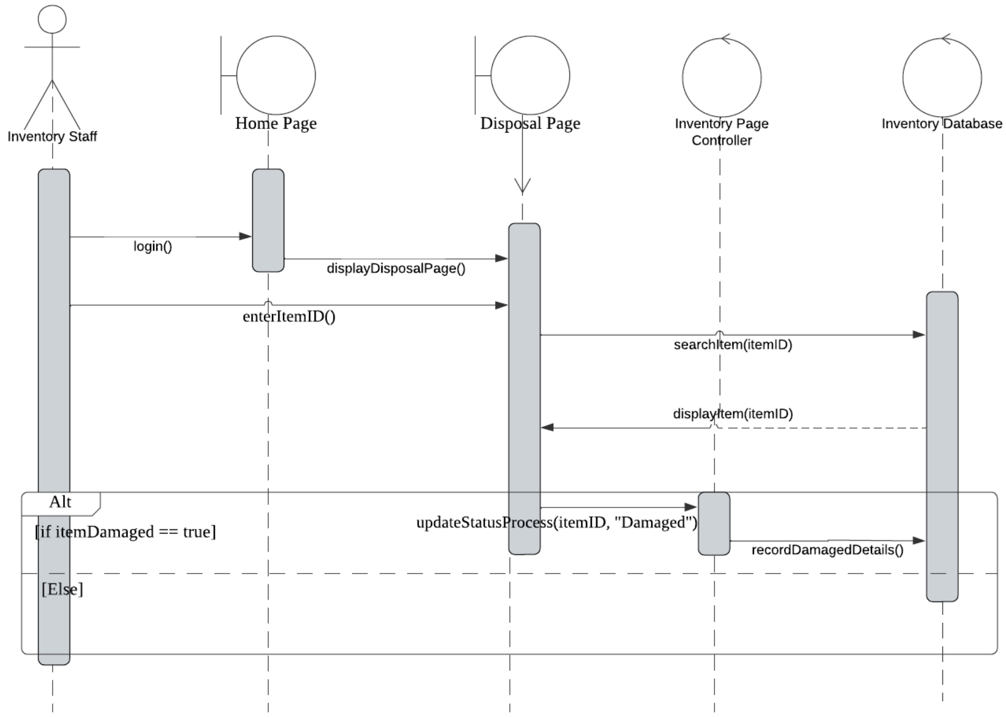


Figure 4.6 Sequence Diagram of Identify Damaged Item

Figure 4.5 and figure 4.6 above shows the workflow starts when the Inventory Staff logs in and searches for an item on the Disposal Page, which retrieves the item's current details from the Database. If the item is identified as damaged, the system updates its status, records the specific details, and saves the changes back to the database. If the item is not damaged, the process simply ends without making any updates.

Verify Item Status

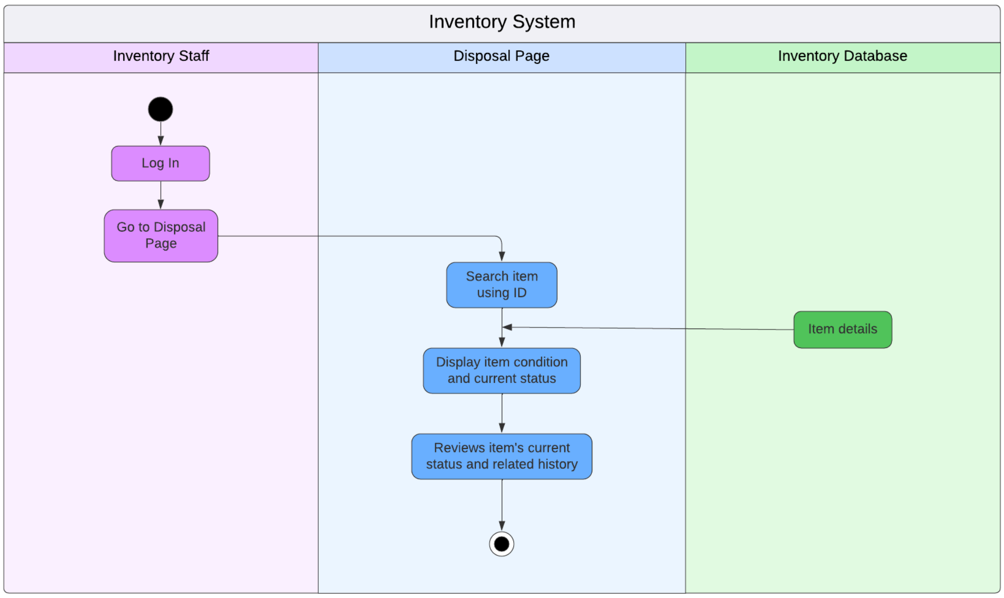


Figure 4.7 Activity Diagram of Verify Damaged Item

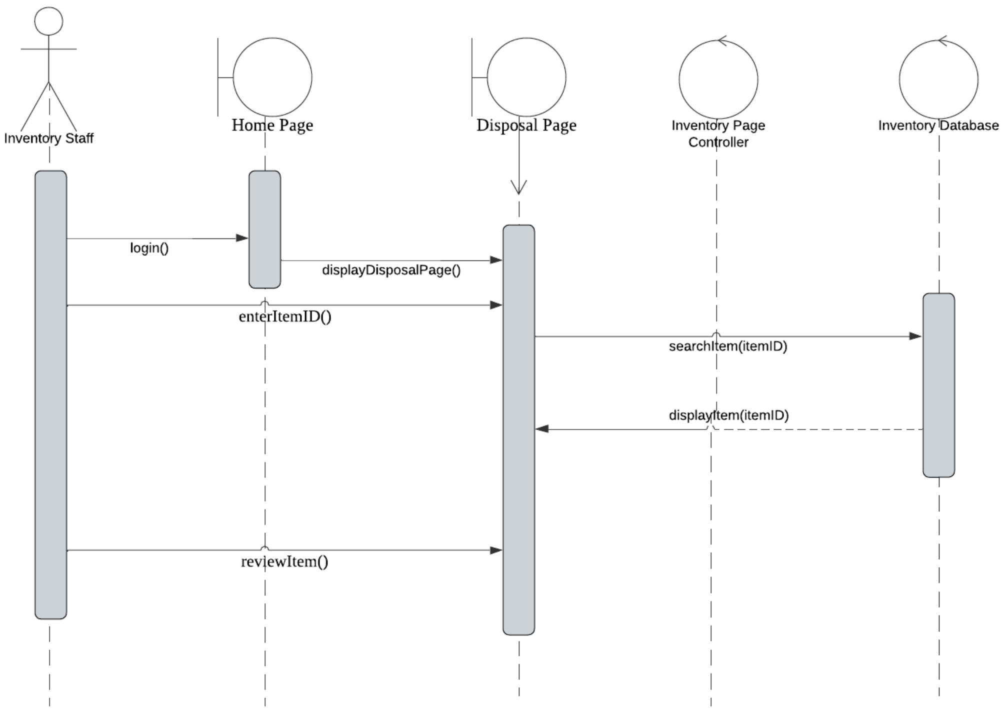


Figure 4.8 Sequence Diagram of Verify Damaged Item

Figure 4.7 and figure 4.8 above is the process starts with the Inventory Staff logging in and navigating to the "Disposal Page." The staff member searches for a specific item using its ID, which triggers the system to retrieve Item details from the Inventory Database. The Disposal Page then displays the item's condition and current status. Finally, the item's current status and related history are reviewed on the page, and the workflow ends.

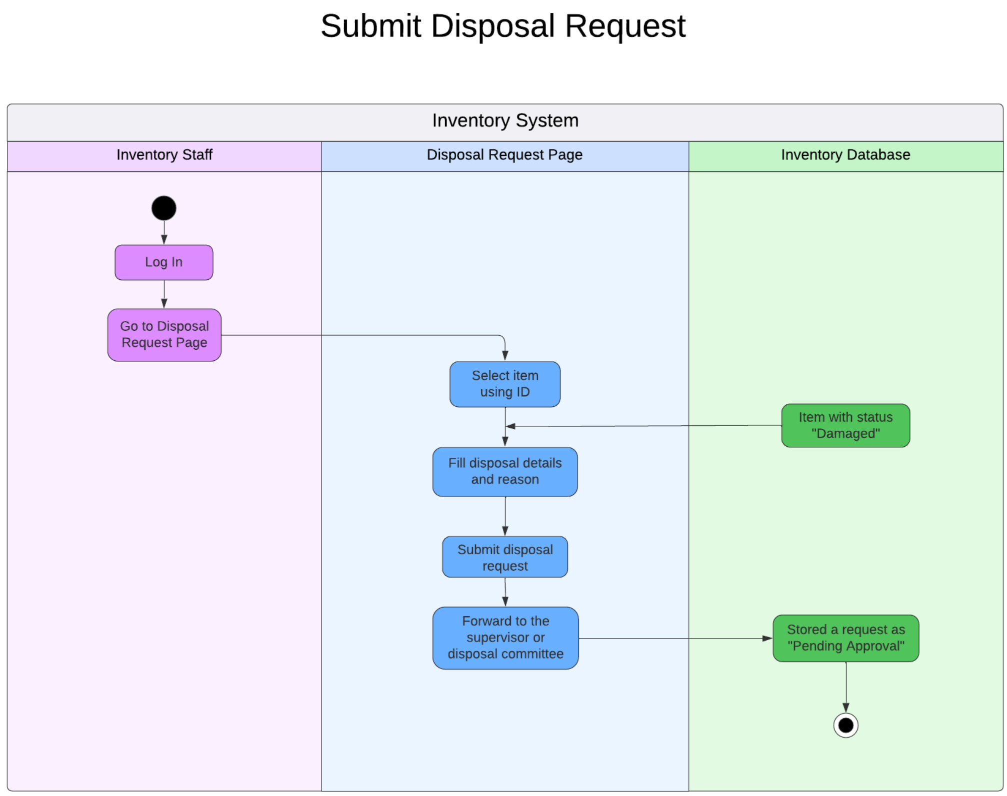


Figure 4.9 Activity Diagram of Submit Disposal Request

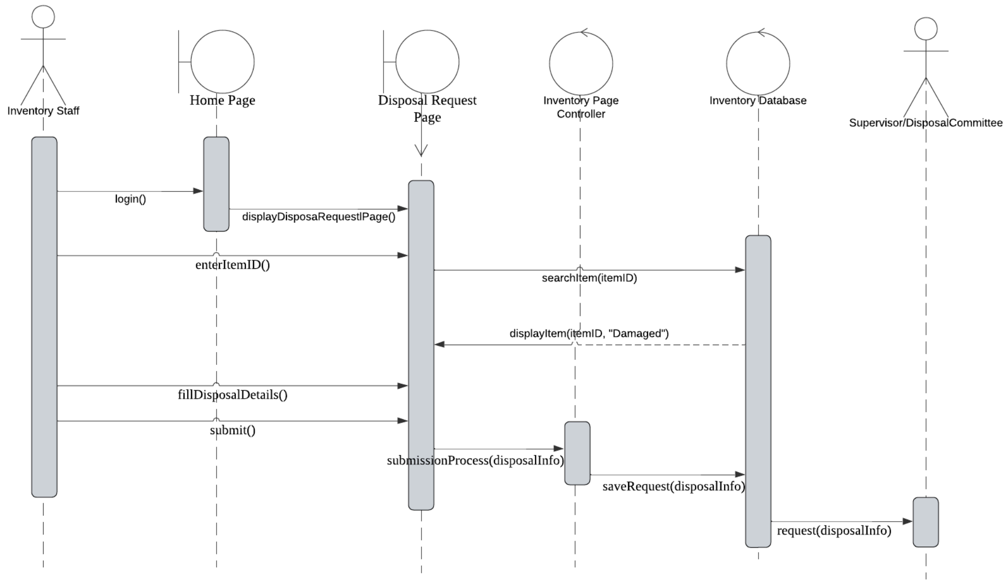


Figure 4.10 Sequence Diagram of Submit Disposal Request

Figure 4.9 and figure 4.10 above shows a workflow that starts with the Inventory Staff selecting a specific item on the Disposal Request Page then the system validates that the item is already marked as "Damaged" in the database. The staff then fills in the disposal details and reason for the request. Upon submission, the system forwards the request to a supervisor/disposal committee and saves it in the Inventory Database as "Pending Approval."

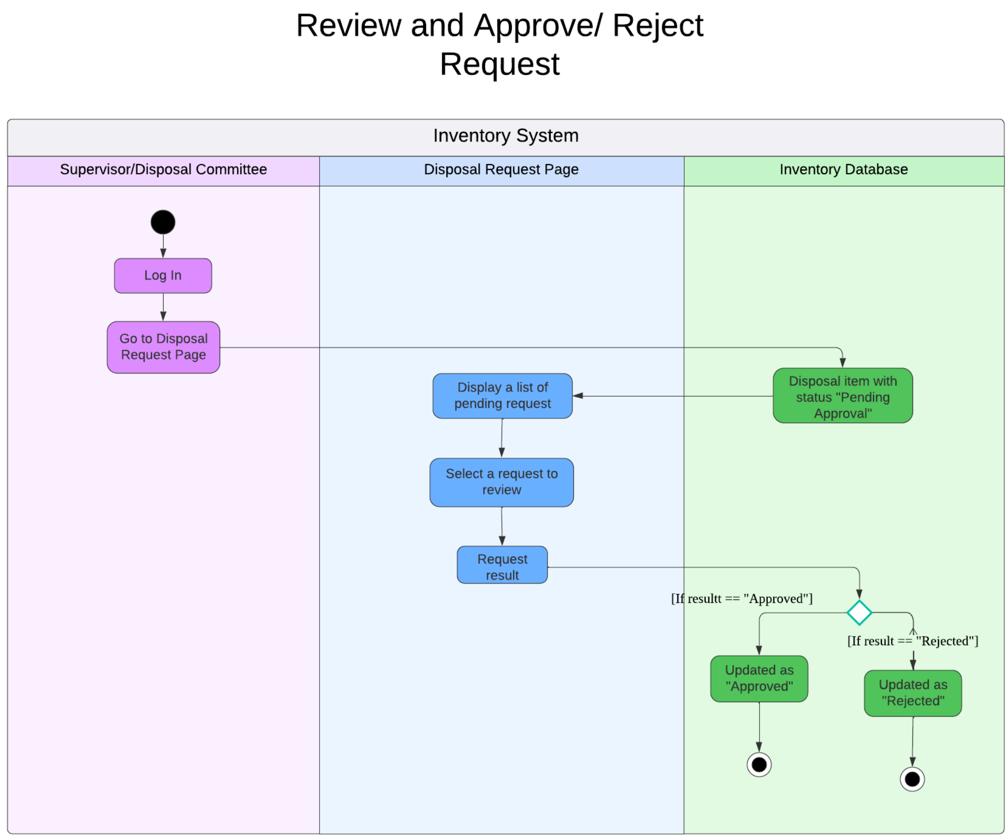


Figure 4.11 Activity Diagram of Review and Approve/Reject Request

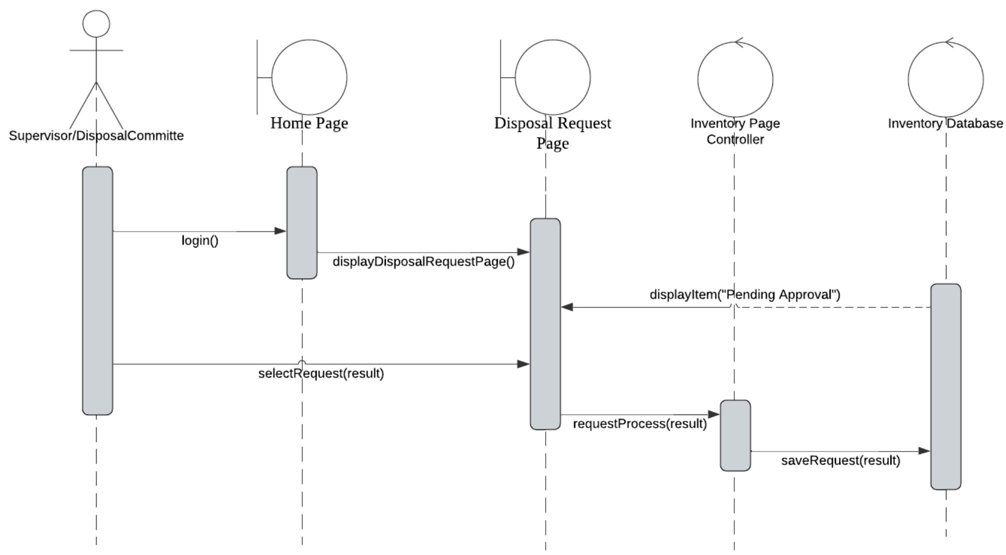


Figure 4.12 Sequence Diagram of Review and Approve/Reject Request

Figure 4.11 and figure 4.12 above shows a workflow that starts with the Supervisor/Disposal Committee navigating to the Disposal Request Page, where the system displays a list of pending requests. The supervisor selects a request to review and either approves or rejects it. Upon approval, the system updates the Inventory Database with the approved disposal item as “Approved” while upon rejection, it marks the request as "Rejected" in the database.

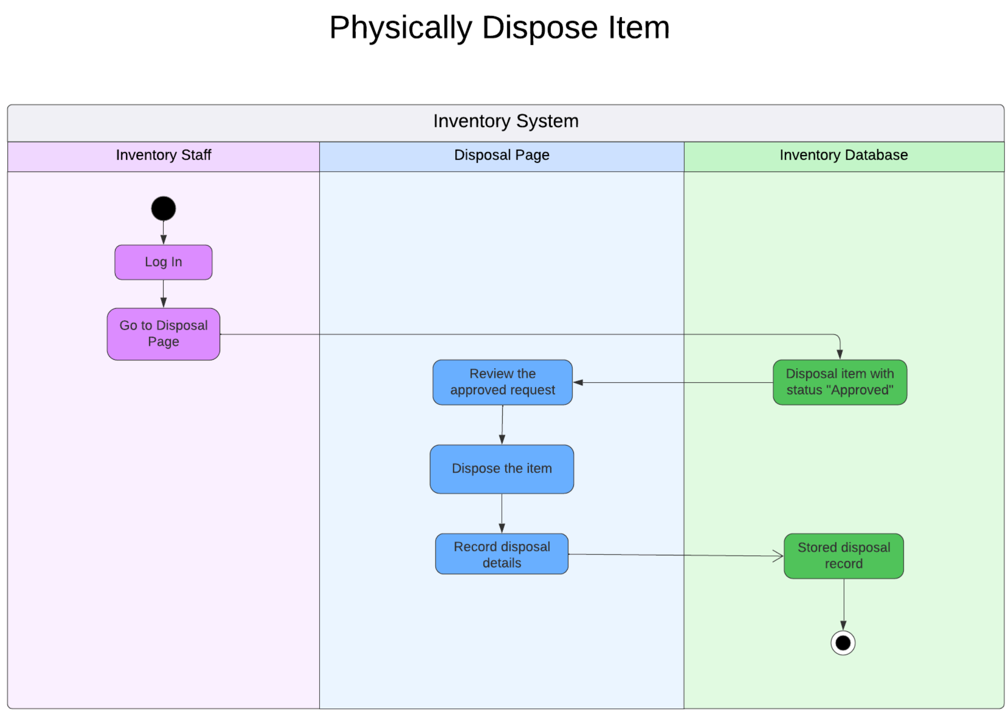


Figure 4.13 Activity Diagram of Physically Dispose Item

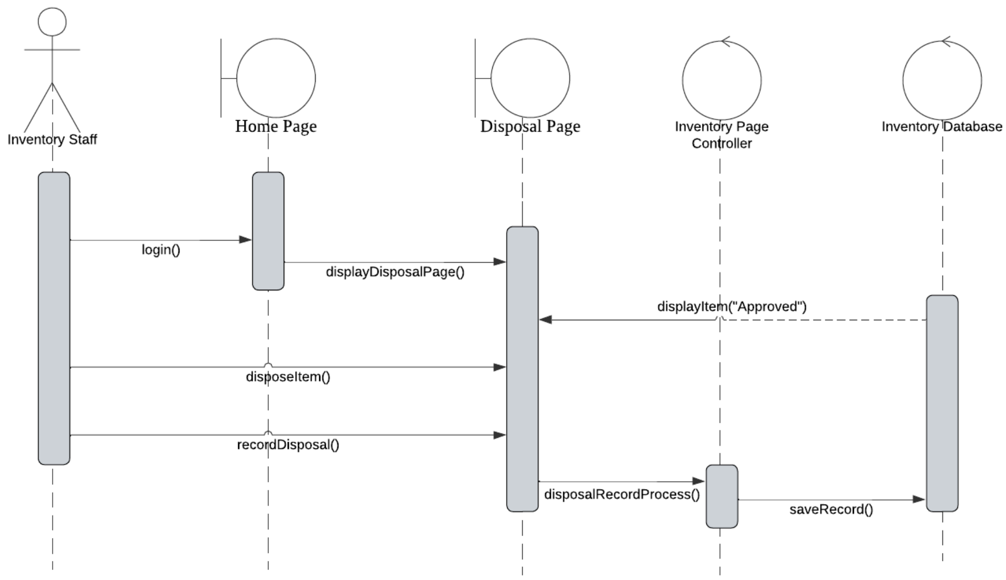


Figure 4.14 Sequence Diagram of Physically Dispose Item

Figure 4.13 and figure 4.14 above shows a workflow that starts with the Inventory Staff logging in and navigating to the Disposal Page to request item disposal. The system then forwards the request for review and approval. Upon supervisor approval, the staff physically disposes of the item, records the disposal details, and the system updates the Inventory Database with the finalized disposal record.

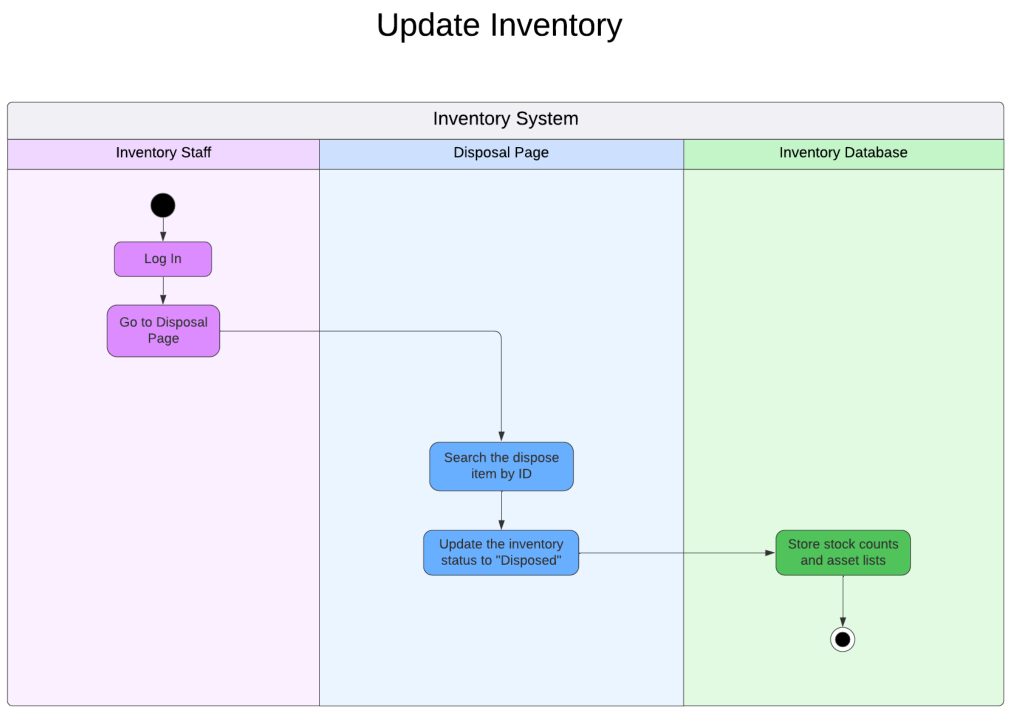


Figure 4.15 Activity Diagram of Update Inventory

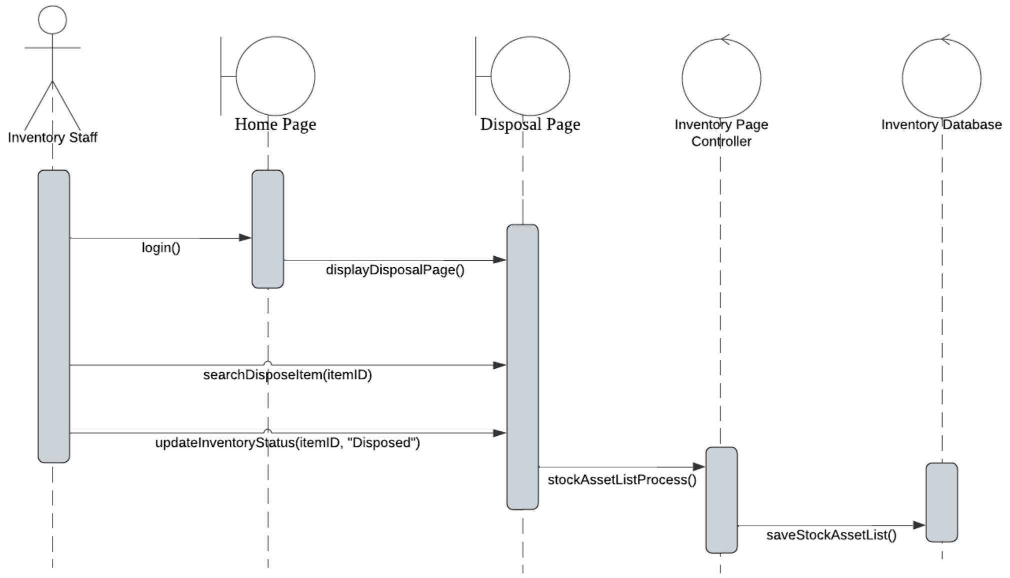


Figure 4.16 Sequence Diagram of Update Inventory

Figure 4.15 and figure 4.16 above shows a workflow that starts with the Inventory Staff logging in and navigating to the Disposal Page, where they search for an item by ID. The system then updates the item's status to "Disposed" in the inventory and adjusts the stock counts accordingly, ensuring the Inventory Database reflects accurate stock levels and asset counts.

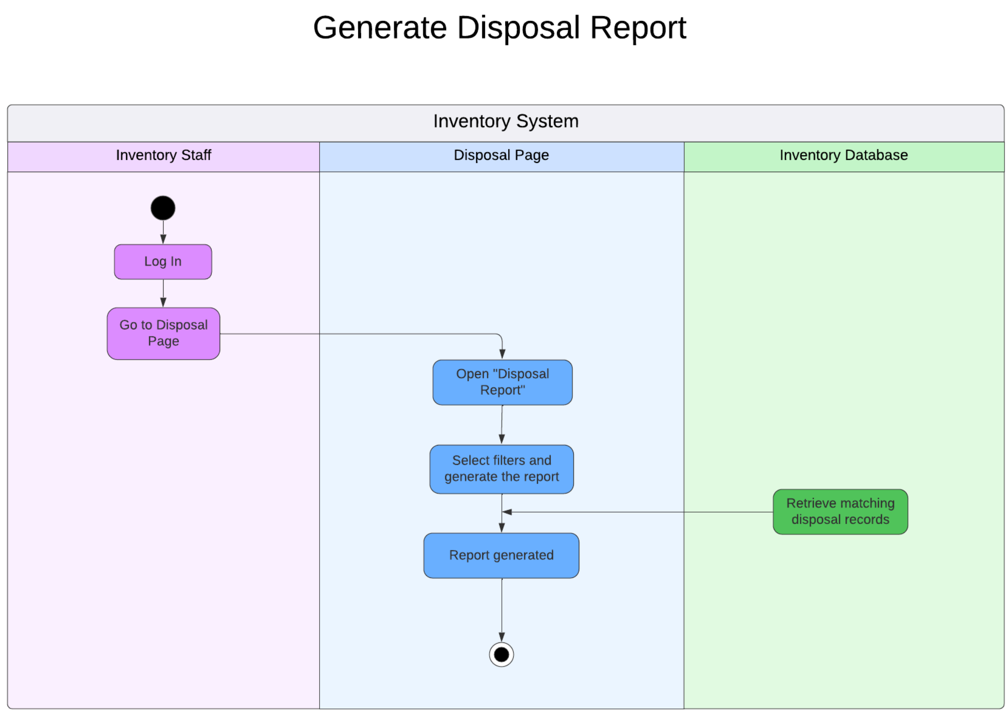


Figure 4.17 Activity Diagram of Generate Disposal Report

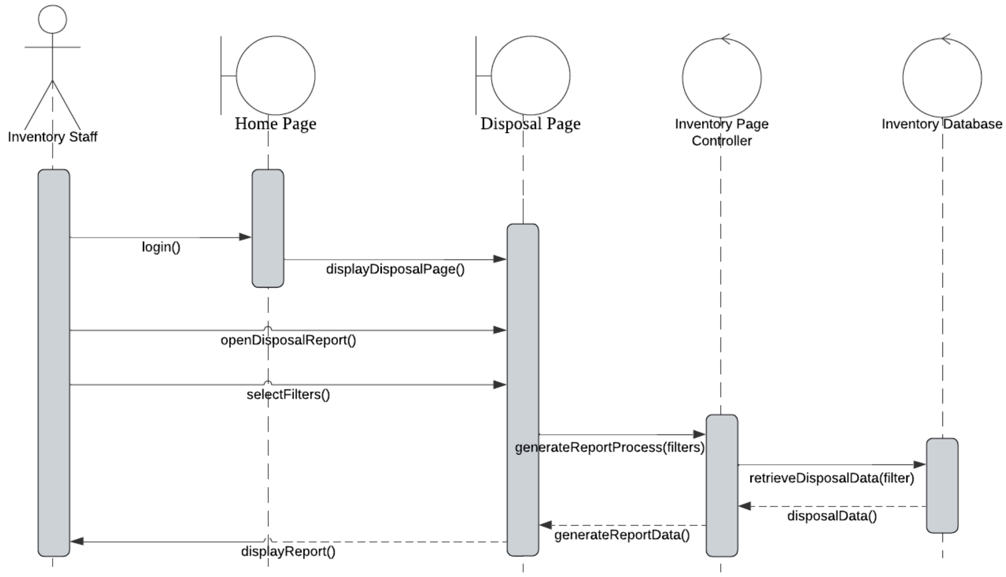


Figure 4.18 Sequence Diagram of Generate Disposal Report

Figure 4.17 and figure 4.18 The process begins when the Inventory Staff opens the "Disposal Report" section and selects specific filters to customize the output. The system then queries the Inventory Database to retrieve matching disposal records based on these filters. Once the relevant data is fetched, the final report is generated and displayed to the user, completing the workflow.

4.7.8 Database Design

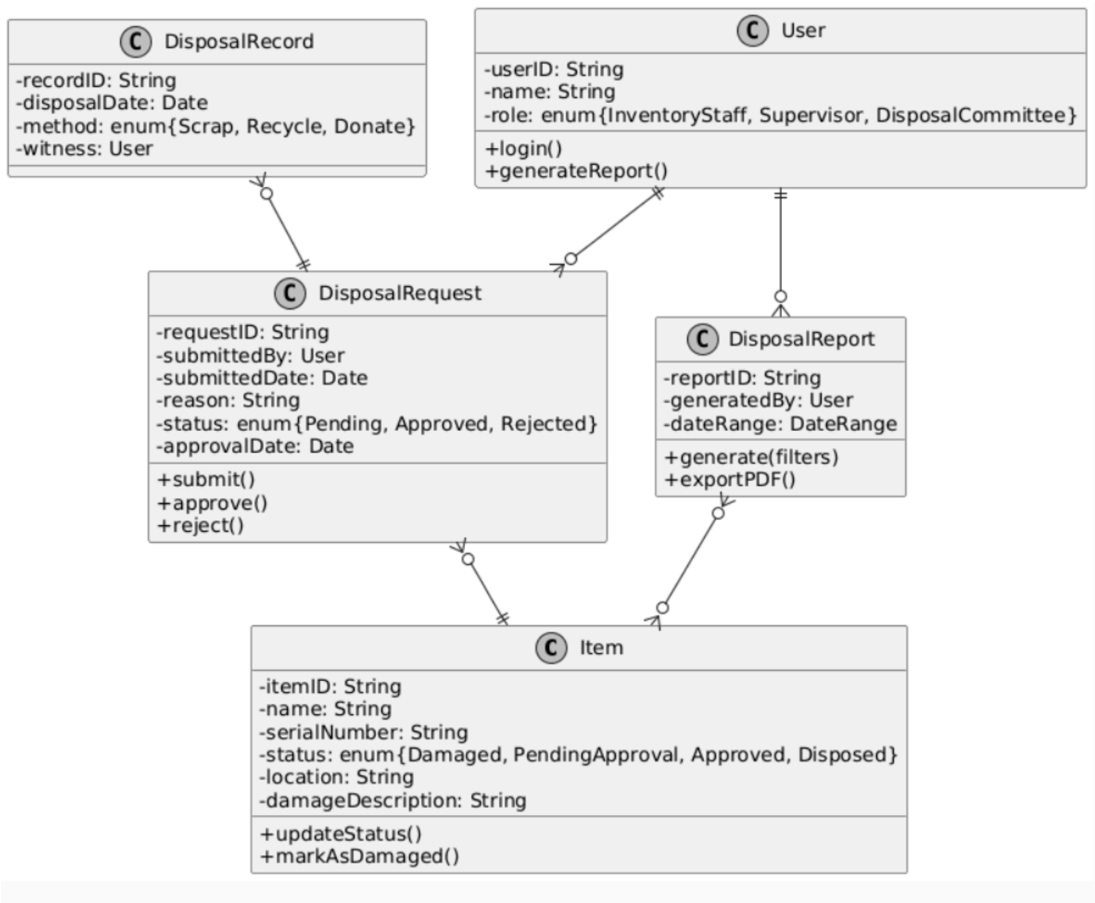


Figure 4.19 Class Diagram of Disposal Inventory System

Figure 4.19 above is the class diagram for this Disposal Inventory System which outlines how the lifecycle of damaged or outdated assets is managed from start to finish with respect to reporting and disposal. The initial step includes capturing the item detail in an Item class. The status of an Item will be tracked based on its condition damaged, under consideration for approval, approved and ultimately disposed of.

The first step after an item has been determined to be damaged is for someone to submit a DisposalRequest to a specific User, such as inventory personnel. A DisposalRequest will contain the reason for the disposal and whether or not approval has been obtained from the appropriate personnel like Supervisor or Disposal Committee.

After the DisposalRequest has received approval, the actual disposal of an item is recorded in the DisposalRecord, which will include the date of disposal, the method of disposal and will name the staff who witnessed the disposal. In addition to recording all disposal activities, staff can also generate and export a DisposalReport that summarizes all disposal activities between a set date range in PDF format. The overall working of this diagram indicates a controlled workflow which provides for transparency and authorization of all disposal activities and provides documentation to support the disposal of items.

4.7.9 Interface Design

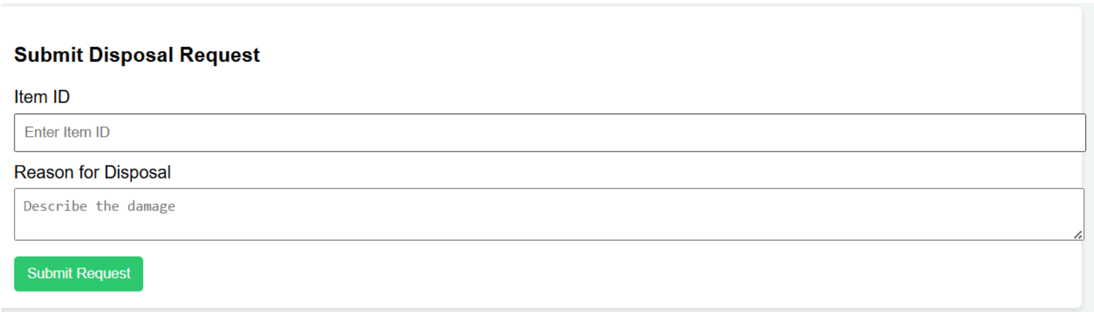


Figure 4.20 Interface of Disposal Request Submission

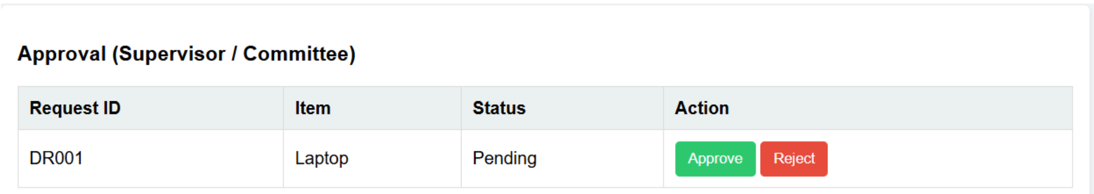


Figure 4.21 Interface of Approval/Rejection from Supervisor or Committee

4.8 Chapter Summary

To summarize, this chapter provided a detailed evaluation of UTHM's organizational hierarchy and current "As-Is" architecture. By synthesizing stakeholder requirements gathered through structured interviews, a "To-Be" Enterprise Architecture was proposed to establish a unified, cloud-based ecosystem featuring centralized data management and Single Sign-On (SSO) integration. To bridge the gap between high-level architectural planning and technical implementation, the project has developed specific design artifacts for the UTHM Inventory Subsystem, including Use Case Diagrams and Descriptions to define actor interactions, Activity Diagrams to map procedural workflows, and Sequence Diagrams to illustrate real-time logic. These behavioral models, combined with comprehensive database and interface designs by utilizing SAP BTP Trial Account, The proposed system also ensures that it is structurally sound, user-centric, and fully prepared for the development phase using SAP BTP technology to support the university's digital transformation.

REFERENCES

Kute, S. S., & Thorat, S. D. (2014). A review on various software development life cycle (SDLC) models. *International Journal of Research in Computer and Communication Technology*, *3*(7), 778-779.

Appendix A

Appendix B

Appendix C